WEATHER APP

Computação Móvel @ Universidade do Porto

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TODAY'S FOCUS

Project Summary

Architecture Overview

External Integrations

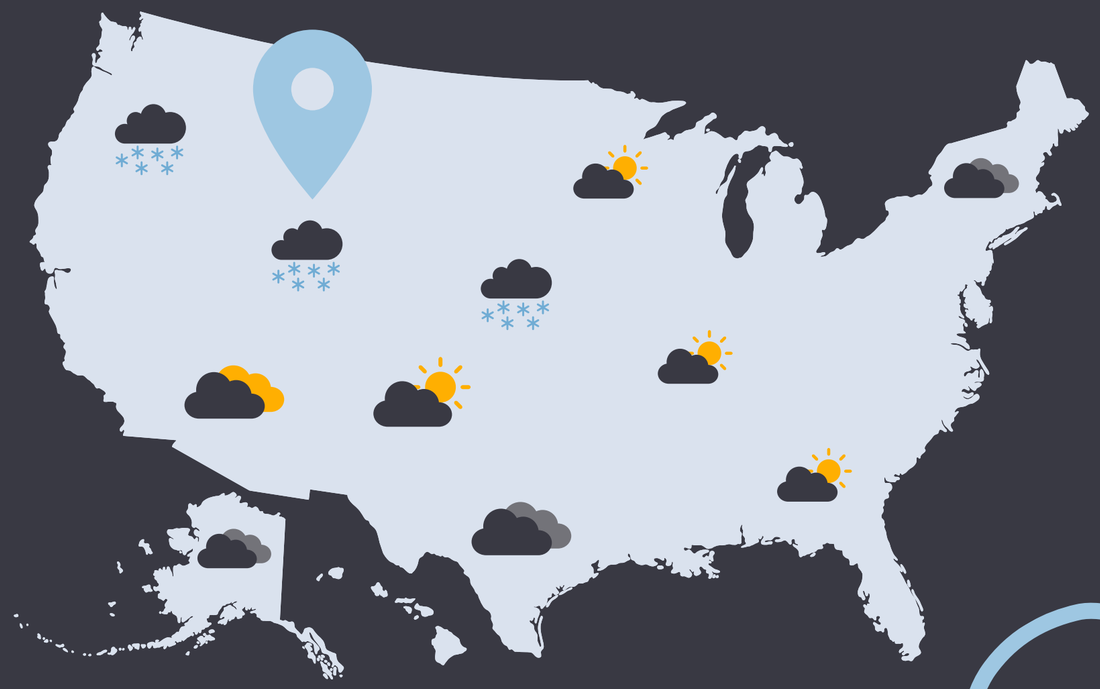
Key Features

Development



Project Summary

- Flutter-based weather app
- Real-time & forecasted weather for multiple cities
- **Features:** hourly forecast, weekly trends, alerts
- Personalized city list and GPS-based current location weather



Key Features

1



Current Weather

Real-time data, icons, alerts

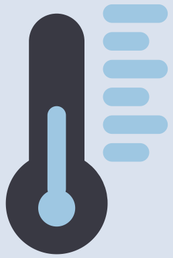
2



Next Days Forecast

Up to 7 days with hourly breakdown

3



Week Stats

Graphs for temperature, humidity, etc.

4



Autocomplete City Search

CSV-based local dataset

Users can select their preferred units for temperature, distance and speed on the **Settings page**. Changes are applied **instantly** across the app in **real-time**.



Architecture Overview

1

Client-Side Only

2

Built with **Flutter**

3

State management using **Provider**

4

Visual Crossing API for weather data

5

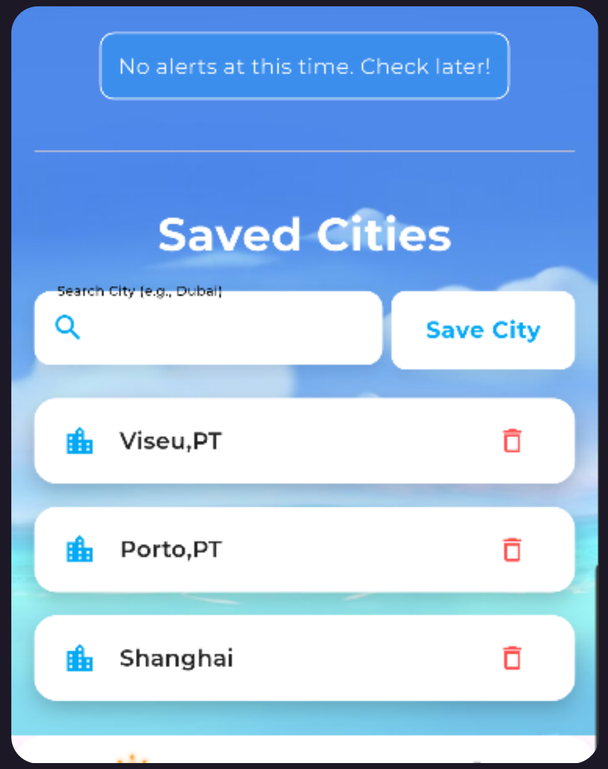
Local storage: sqflite for cities and settings

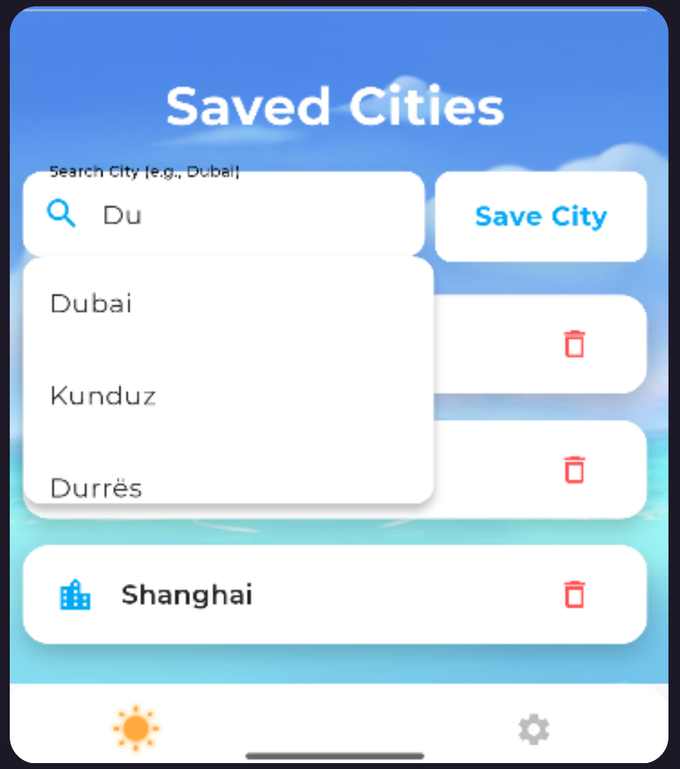
API Request

$baseUrl/{$city}/{$date}?
include=days,hours,current&key=API_KEY&iconSet=icons2;

City Management

- Add cities with a system of auto completion and results suggestions.
- Remove cities at any time.
- Access weather data for saved cities.



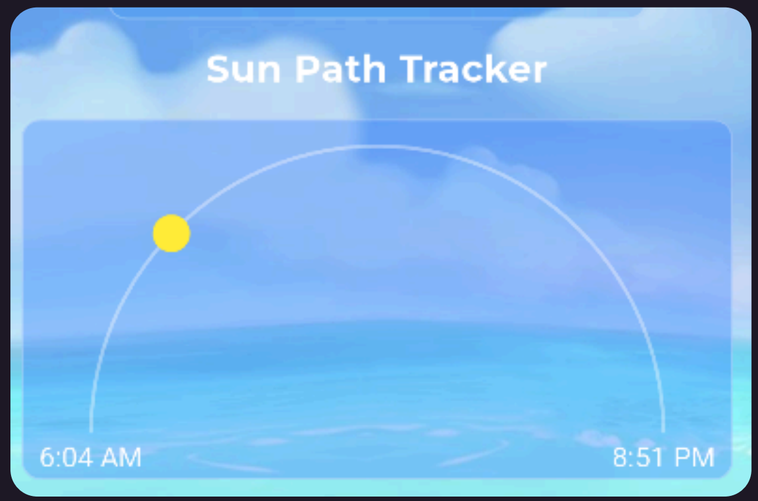


City Weather View (3 Tabs)

1st Tab (Current Weather):

- Temp, humidity, wind, pressure, etc. with weather icon, weather alerts for that day and a visual demonstration of when the sun rises and sets.



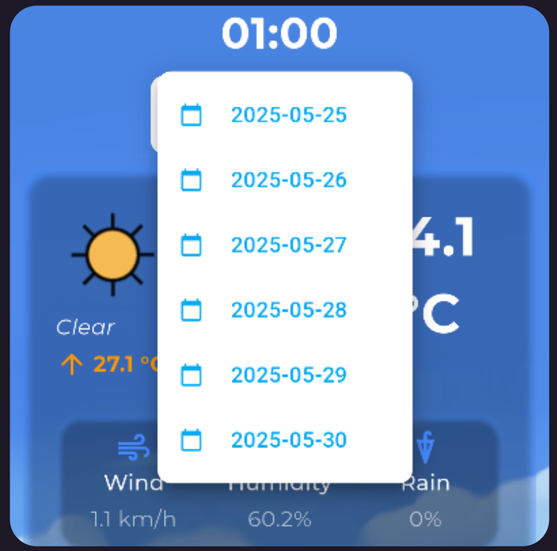


City Weather View (3 Tabs)

2nd Tab (Next Days Forecast):

- By default, the next day is selected, however, the user can select another of the 7 following days.
- Includes a summary with hourly breakdown and corresponding icons and weather alerts for that day.





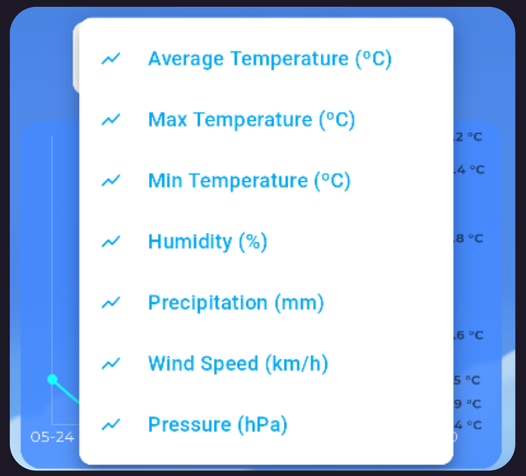
City Weather View (3 Tabs)

3rd Tab (Week Stats):

- Graphical demonstration of the weather's week evolution.
- User can select a time interval and pick a weather statistic to study.

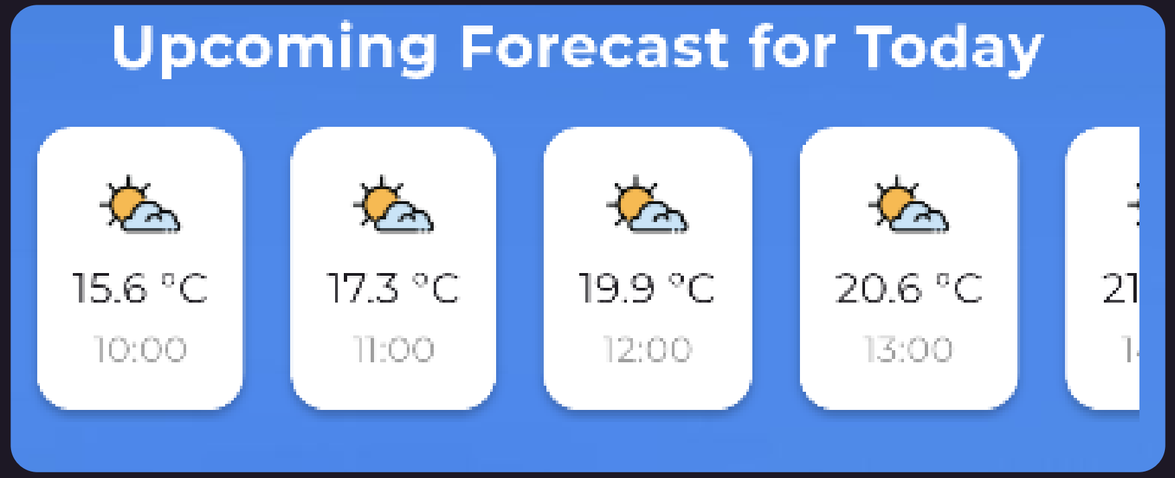






Hourly Forecast

- View hourly weather for the selected day.
- Clicking a tile updates the weather card to display detailed information for the selected timestamp.



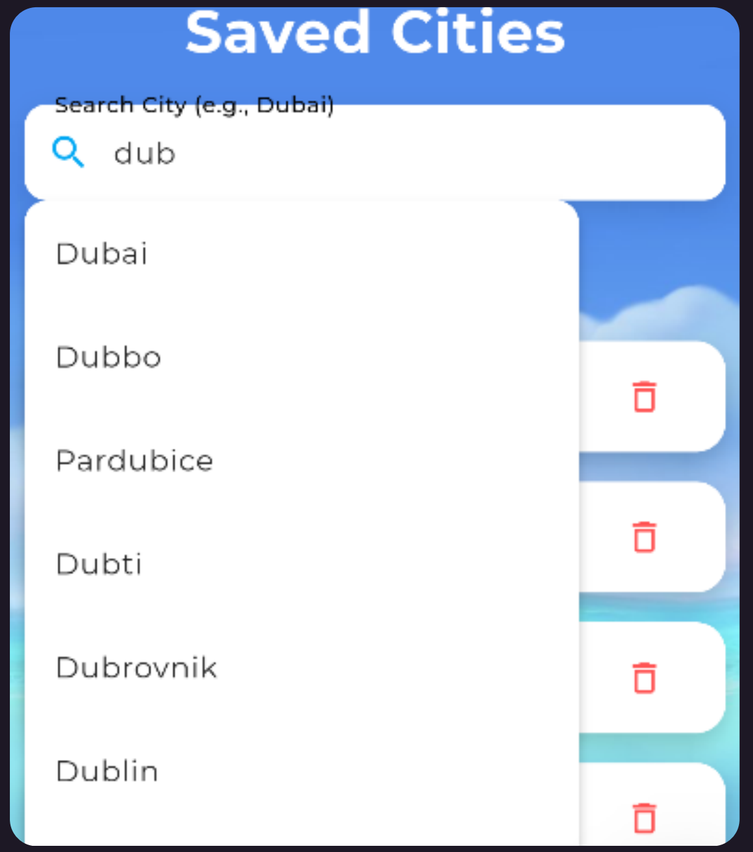
Current Location Weather

- Uses the device’s GPS to automatically detect the user’s current location.
- Geolocation data is then used to request weather information from the API.



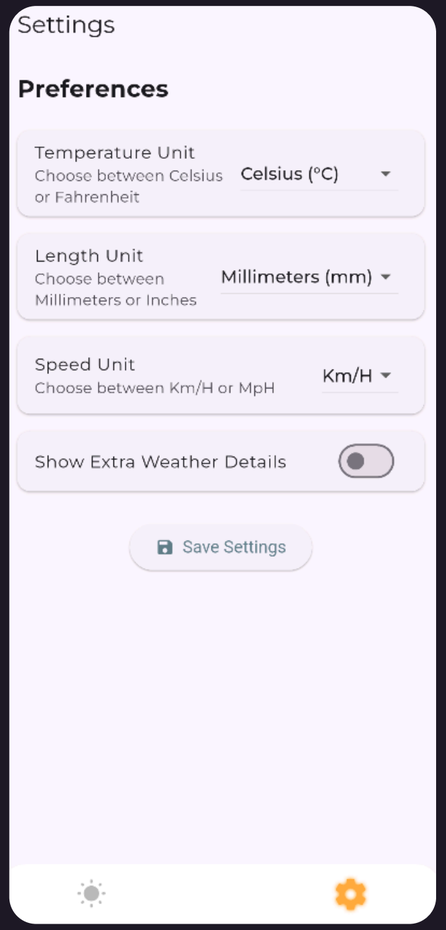
Autocomplete mechanism

- When users begin typing to search for and save a city, a dropdown menu appears with dynamic suggestions based on their input.
- Tried using API's like **GeoDB Cities** and Google's **Places API** but opted for an offline solution.
- Used **csv** dataset and **Autocomplete** widget.



Settings Page

- User can pick preferred units:
  - Temperature
  - Distance
  - Speed
- Can also choose to already activate the switch in the weather card, used to display additional weather information.



Local Database Tables

CITIES		
Column	**Type**	**Description**
id	INTEGER	Primary key, auto-incremented
name	TEXT	City name (unique)

SETTINGS		
Column	**Type**	**Description**
key	TEXT	Setting name (PK)
value	TEXT	Value of the setting

External Integrations

Weather API

Retrieve weather data

http

API requests

Geolocator/Geocoding

GPS & reverse geolocation

sqflite

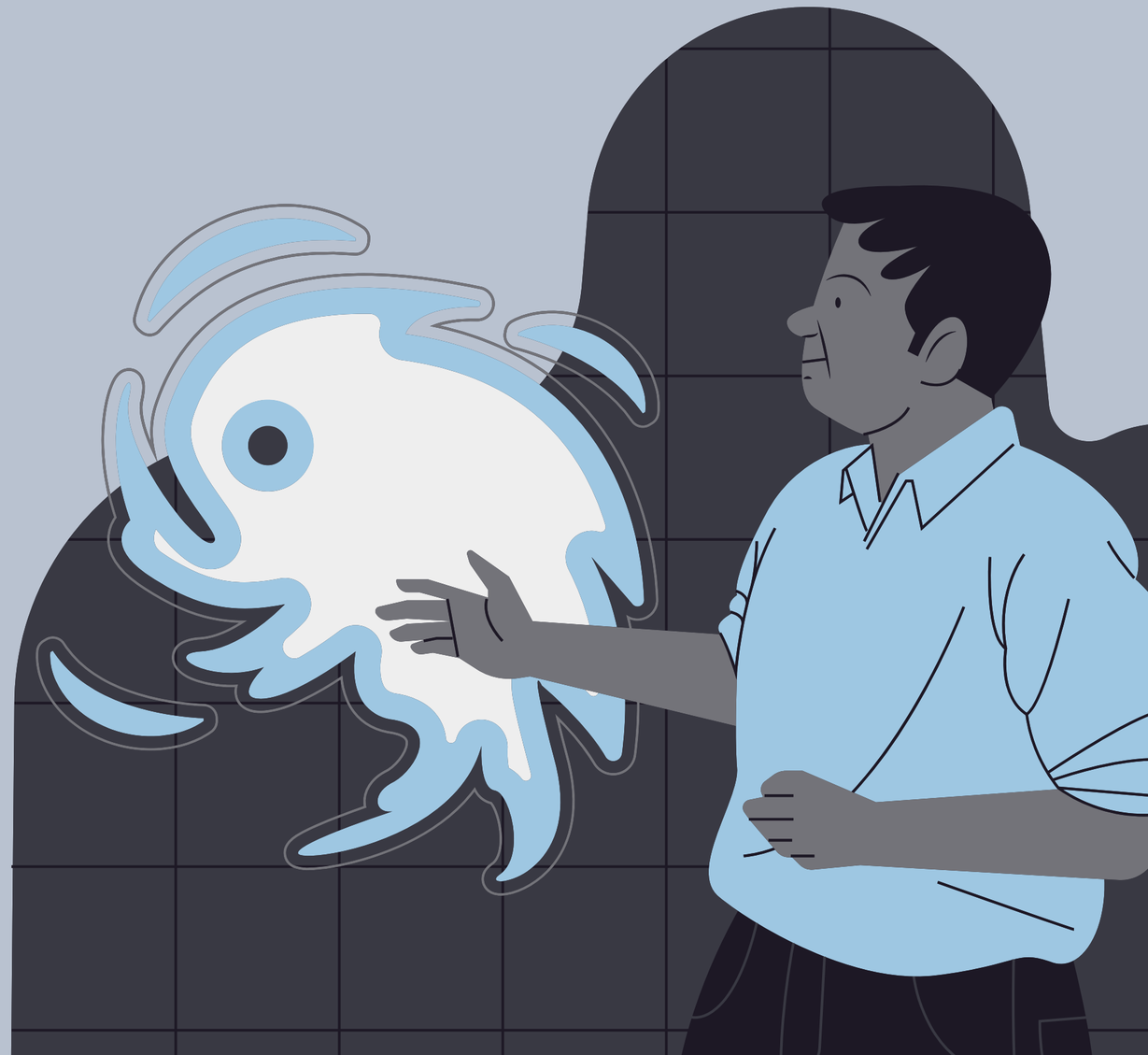
Local Storage

fl_chart

Graph rendering

csv

City autocomplete functionality



Thanks for listening!

Any Questions?
We'll answer them!

